SOFTWARE ENGINEERING LAB – 2

Agile Backlog Creation & Sprint Simulation in Jira

Project: Automated Rubric Assignment Evaluator

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1. Project Overview

This lab converts the functional requirements from Lab 1 into an Agile backlog, organizes them into Epics and User Stories, assigns priorities and Fibonacci story points, and simulates two Scrum sprints in Jira.

Project Title: Automated Rubric Assignment Evaluator

Primary Domain: Campus & Academic Operations

Target Actors: Student, Faculty Evaluator, Peer Reviewer, Notification Service

The Lab 1 requirements define automated project submission and evaluation, rubric configuration, peer-review assignment, faculty score overrides, and notifications. These functions form the basis of the Jira backlog.

2. Epics

Epic	Description	Related Lab 1 Requirements
Epic 1: Submission & Automated Evaluation	Handles student project submission, queueing, automated test execution, and generation of rubric scores.	FR-001
Epic 2: Rubric Configuration	Allows faculty evaluators to configure rubric criteria, weights, and associated unit test suites.	FR-002
Epic 3: Peer Review Management	Automates assignment of peer reviewers and supports submission of peer reviews.	FR-003
Epic 4: Score Review & Notifications	Allows faculty to override or annotate scores and notifies students and reviewers when evaluation activities are complete.	FR-004, FR-005

3. User Stories, Priorities & Story Points

The user stories use the required Agile format: As a [user role], I want [goal], so that [benefit]. Story points use the Fibonacci-style scale described in the lab handout.

Epic	User Story	Description	Priority	Points
Submission & Automated Evaluation	**Upload Project Submission**	As a student, I want to upload my project as a ZIP file, so that the system can evaluate my submission automatically.	High	3
Submission & Automated Evaluation	**Execute Automated Unit Tests**	As a student, I want my submitted project to be automatically tested, so that my implementation can be evaluated objectively.	High	5
Submission & Automated Evaluation	**Generate Rubric Score Breakdown**	As a student, I want to receive an itemized rubric score based on test results, so that I can understand how my grade was derived.	High	5
Rubric Configuration	**Configure Assignment Rubric**	As a faculty evaluator, I want to configure rubric categories and weights, so that each assignment can be evaluated consistently.	High	5
Rubric Configuration	**Associate Unit Test Suite**	As a faculty evaluator, I want to associate a unit test suite with an assignment, so that student submissions can be auto-tested.	High	3
Peer Review Management	**Assign Peer Reviewers**	As a faculty evaluator, I want the system to automatically assign peer reviewers to submissions, so that peer feedback is distributed fairly.	Medium	5
Peer Review Management	**Submit Peer Review**	As a peer reviewer, I want to submit feedback for an assigned project, so that the student can receive additional insight.	Medium	3
Score Review & Notifications	**Manually Override Rubric Score**	As a faculty evaluator, I want to manually override or annotate an automated score, so that exceptional cases are handled fairly.	High	5
Score Review & Notifications	**Notify Student of Evaluation**	As a student, I want to receive a notification when my evaluation is complete, so that I know when my results are ready.	Medium	3
Score Review & Notifications	**Notify Peer Reviewers**	As a peer reviewer, I want to receive a notification when I am assigned a submission, so that I know when I need to review.	Medium	3

Total planned backlog: 40 story points.

4. Jira Backlog Screenshot

Screenshot showing the Epics, User Stories, priorities, and story points in Jira.

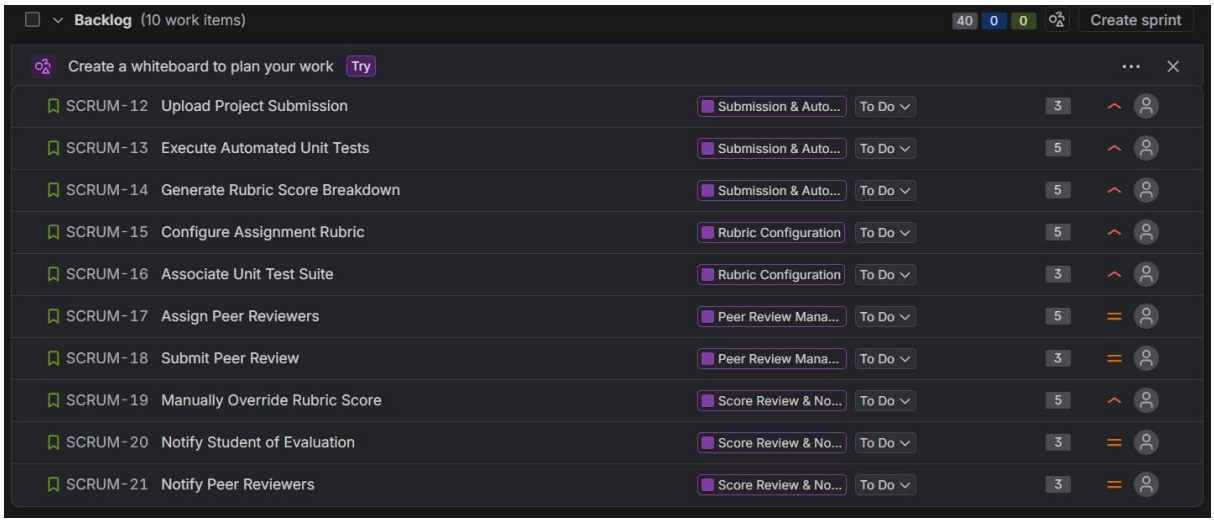


Figure 1: Jira backlog with Epics, priorities, and story points.

5. Sprint 1 – Active Sprint Screenshot

Jira Active Sprint screenshot showing stories moving through To Do, In Progress, and Done.

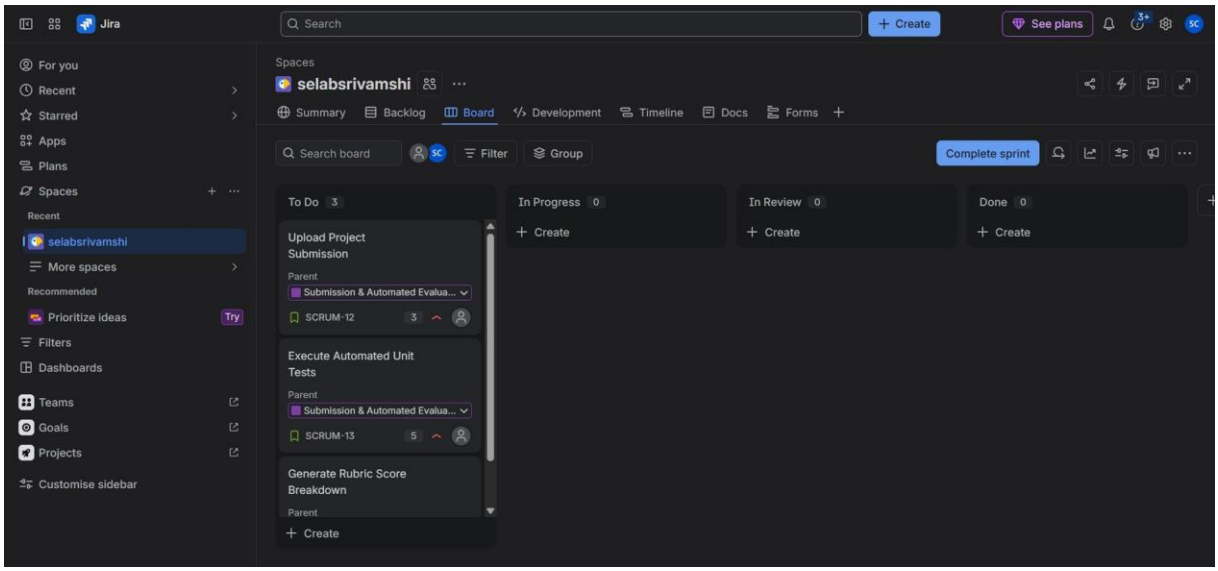


Figure 2: Sprint 1 active sprint board.

6. Sprint 2 – Active Sprint Screenshot

Jira Active Sprint screenshot for the second sprint.

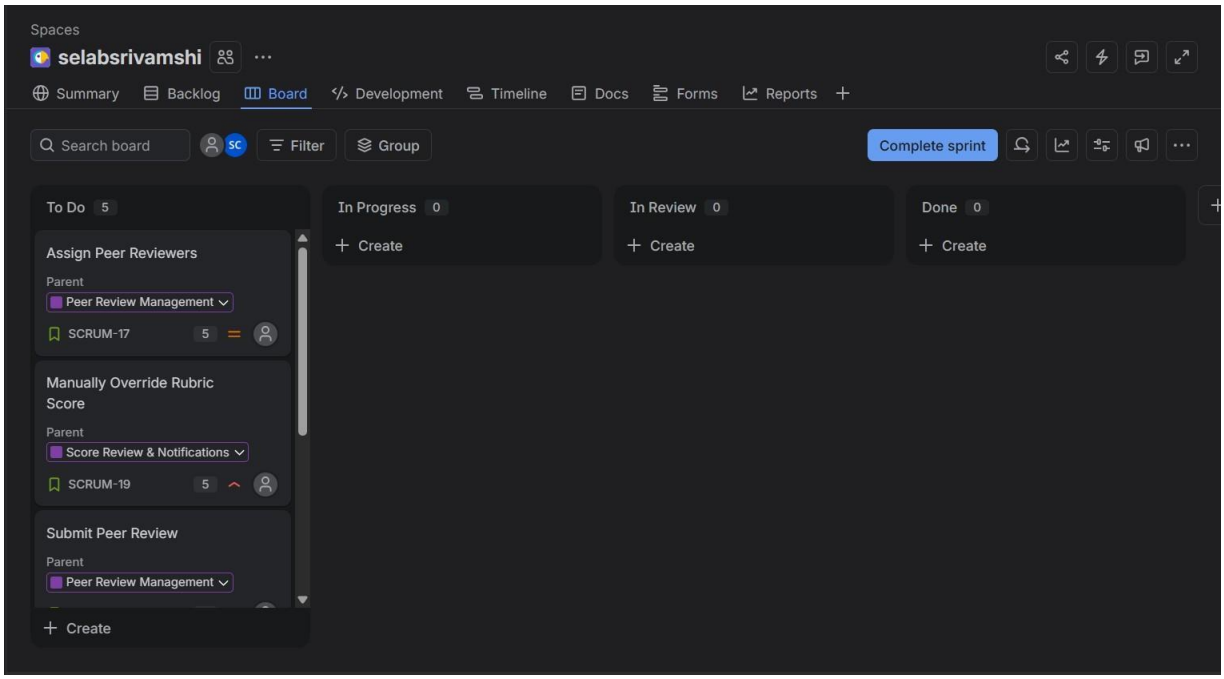


Figure 3: Sprint 2 active sprint board.

7. Sprint Plan

Sprint 1 is planned as a one-week sprint containing the core automated evaluation workflow. Sprint 2 contains peer-review, score-review, and notification functionality.

Sprint 1

User Story	Story Points
Upload Project Submission	3
Execute Automated Unit Tests	5
Generate Rubric Score Breakdown	5
Configure Assignment Rubric	5
Associate Unit Test Suite	3
Total	**21**

Sprint 2

User Story	Story Points
Assign Peer Reviewers	5
Submit Peer Review	3
Manually Override Rubric Score	5
Notify Student of Evaluation	3
Notify Peer Reviewers	3
Total	**19**

8. Burndown Chart

Burndown Chart from Jira after running the sprint(s), showing remaining work and the expected progress guideline.

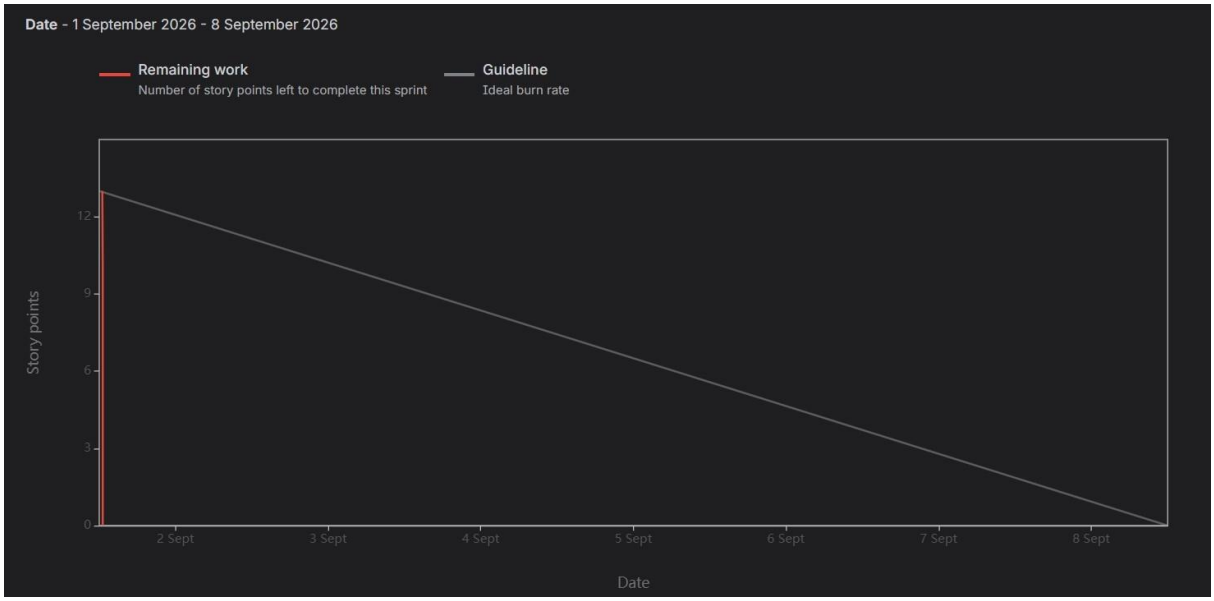


Figure 4: Sprint burndown chart.

9. Reflection Questions

1. Did your estimations reflect the actual effort?

Yes, the story point estimates generally reflected the effort required. Smaller stories such as uploading submissions and sending notifications required fewer points, while automated testing, score generation, and score overrides were more complex and received higher estimates.

2. Was your backlog well-prioritized?

Yes. High-priority stories related to the core evaluation workflow were placed before medium-priority peer-review and notification features. This ensured that the essential evaluation functionality was addressed first.

3. How did your simulated sprint align with your plan?

The simulated sprints followed the planned backlog. Sprint 1 focused on submission, rubric configuration, automated testing, and score generation. Sprint 2 focused on peer review, score review, and notifications.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed how quickly the planned story points were completed during the sprint. It provided an indication of whether progress matched the planned workload and whether the amount of work committed to the sprint was realistic.

10. Submission Checklist

Item	Status
Epics created in Jira	Completed
User Stories created and linked to Epics	Completed
Priorities assigned	Completed
Story Points assigned	Completed
Sprint 1 completed	Completed
Sprint 2 completed	Completed
Active Sprint screenshot added	Completed
Burndown Chart screenshot added	Completed
Reflection questions answered	Completed